



兰州工业学院

实 验 报 告

(2022 — 2023 学年第 二 学期)

课程名称 《计算机网络应用》

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实验项目	实验二 VLAN 间通信		
实验时间	2023 年 4 月 8 日	实验地点	信息与通信工程基础实验室 (EDA 技术实验室)

一、实验目的

- 1.理解虚拟局域网的功能与作用；
- 2.掌握交换机的基本配置命令；
- 3.基于端口的 VLAN 划分方法；
- 4.掌握使用三层交换机实现不同 VLAN 间的通信。

二、实验环境：

Cisco Packet Tracer 模拟软件

三、实验步骤

1、 在一台 2960 交换机上设置 VLAN

任务要求：

- (1)在一台已经进行初始设置的 2960 交换机上设置两个 VLAN: VLAN10、VLAN20，分别取名为 students、teachers。
- (2) PC1 和 PC2 是 students 中的成员，PC3 是 teachers 中的成员。
- (3) 验证设置前后 PC1、PC2、PC3 是否能够 ping 通。

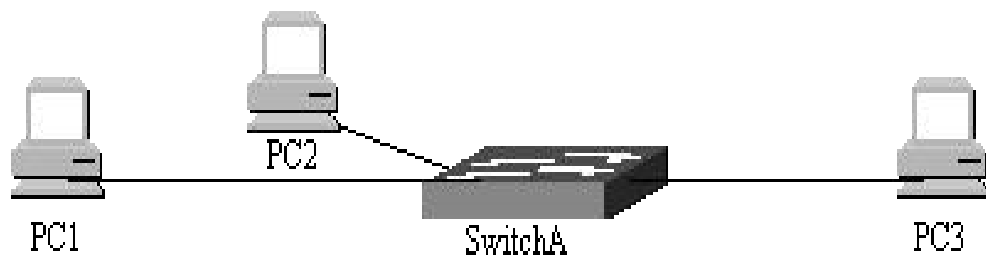
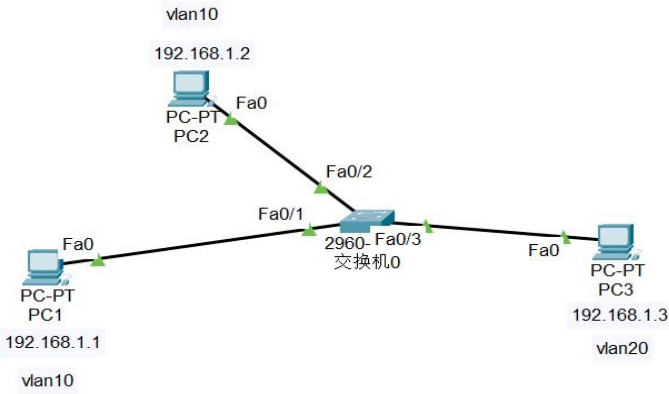


图 1 实验拓扑结构图

在 Cisco Packet Tracer 模拟软件的拓扑图

	
PC1、PC2、PC3 和交换机连接的端口信息	<pre>! interface FastEthernet0/1 switchport access vlan 10 ! interface FastEthernet0/2 switchport access vlan 10 ! interface FastEthernet0/3 switchport access vlan 20 !</pre>
PC1、PC2、PC3 的 IP 地址、子网掩码及网关信息	PC1 192.168.1.1, 255.255.255.0 PC2 192.168.1.2, 255.255.255.0 PC3 192.168.1.3, 255.255.255.0
配置前 PC 之间的连通性测试结果图	

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

```

图 1 PC1 ping 192.168.1.2 和 1.3 发现能够 ping 通

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

图 2 PC2 ping 192.168.1.1 和 1.3 发现能够 ping 通

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<lms TTL=128
Reply from 192.168.1.1: bytes=32 time<lms TTL=128
Reply from 192.168.1.1: bytes=32 time<lms TTL=128
Reply from 192.168.1.1: bytes=32 time<lms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<lms TTL=128
Reply from 192.168.1.2: bytes=32 time<lms TTL=128
Reply from 192.168.1.2: bytes=32 time<lms TTL=128
Reply from 192.168.1.2: bytes=32 time<lms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|

```

图 3 PC3 ping 192.168.1.1 和 1.2 能够

VLAN 配置命令

```

Switch>ena
Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name students
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name teachers
Switch(config-vlan)#exit
Switch(config)#inte
Switch(config)#interface fa
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#switch
Switch(config-if)#switchport acc
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/2
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/3
Switch(config-if)#switchport access vlan 20
Switch(config-if)#

```

图 4 在交换机配置 vlan，设置了 vlan10 和 vlan20 两个 vlan 口，分别命名 students、teachers，将连接的 f 端口将入对应的 vlan 口

配置 VLAN 后 PC 之间的连通性测试结果图

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

图 5 配置 vlan 之后发现 PC1 能够 ping 通同一 vlan 口的 PC2，不同 valn 口的 PC3 ping 通不了

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

图 6 配置 vlan 之后发现 PC2 能够 ping 通同一 vlan 口的 PC1，不同 valn 口的 PC3 ping 通不了

```

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

图 7 发现和其他 vlan 口的两个 PC ping 通不了

2、在一台 3560 三层交换机上实现不同 VLAN 间的通信。

任务要求：

(1) 在一台已经进行初始设置的 2960 交换机上设置三个 VLAN：VLAN10、VLAN20、VLAN30。

(2) PC0 是 VLAN10 中的成员，PC1 是 VLAN20 中的成员，PC2 是 VLAN30 中的成员。

(3) 验证设置前后 PC1、PC2、PC3 是否能够 ping 通。

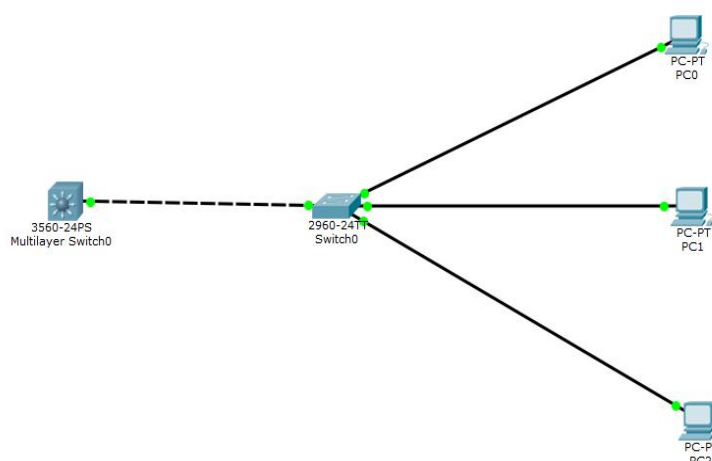
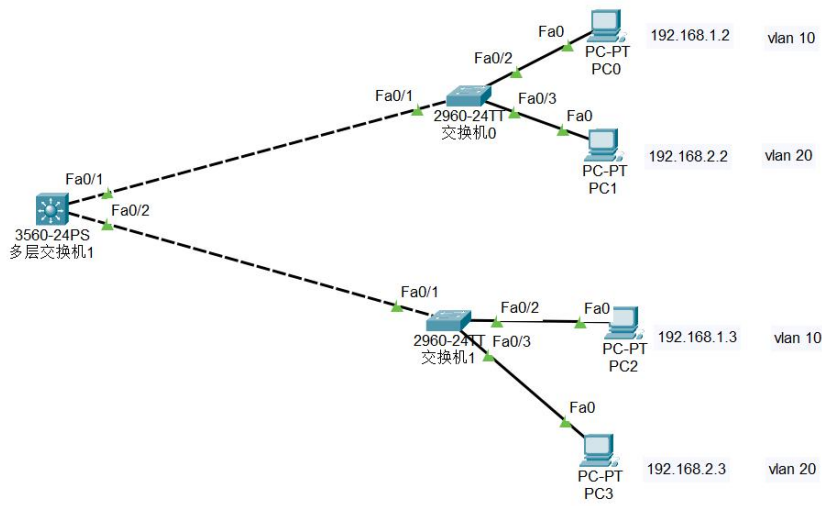


图 1 实验拓扑结构图

在 Cisco Packet Tracer 模拟软件的拓扑图

	
PC1、PC2、PC3 与交换机 2960 连接的端口信息	<pre>interface FastEthernet0/2 switchport access vlan 10 ! interface FastEthernet0/3 switchport access vlan 20 !</pre>
交换机 2960 与交换机 3560 连接的端口信息	<pre>interface FastEthernet0/1 switchport mode trunk ! interface FastEthernet0/1 switchport trunk encapsulation dot1q switchport mode trunk ! interface FastEthernet0/2 switchport trunk encapsulation dot1q switchport mode trunk !</pre>
PC1、PC2、PC3 的 IP 地址、子网掩码及网关信息	PC0:192.168.1.2, 255.255.255.0 PC2:192.168.1.3, 255.255.255.0 网关: 192.168.1.1 PC1:192.168.2.2, 255.255.255.0 PC3:192.168.2.3, 255.255.255.0 网关: 192.168.2.1
未进行交换机 3560 配置前 PC 之间的连通性测试结果图	

```

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<lms TTL=128
Reply from 192.168.1.3: bytes=32 time<lms TTL=128
Reply from 192.168.1.3: bytes=32 time<lms TTL=128
Reply from 192.168.1.3: bytes=32 time<lms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>

```

图 8 PC0 ping 192.168.1.3, 192.168.2.2 发现同一网段的能够 ping 通, 不同网段的无法 ping 通

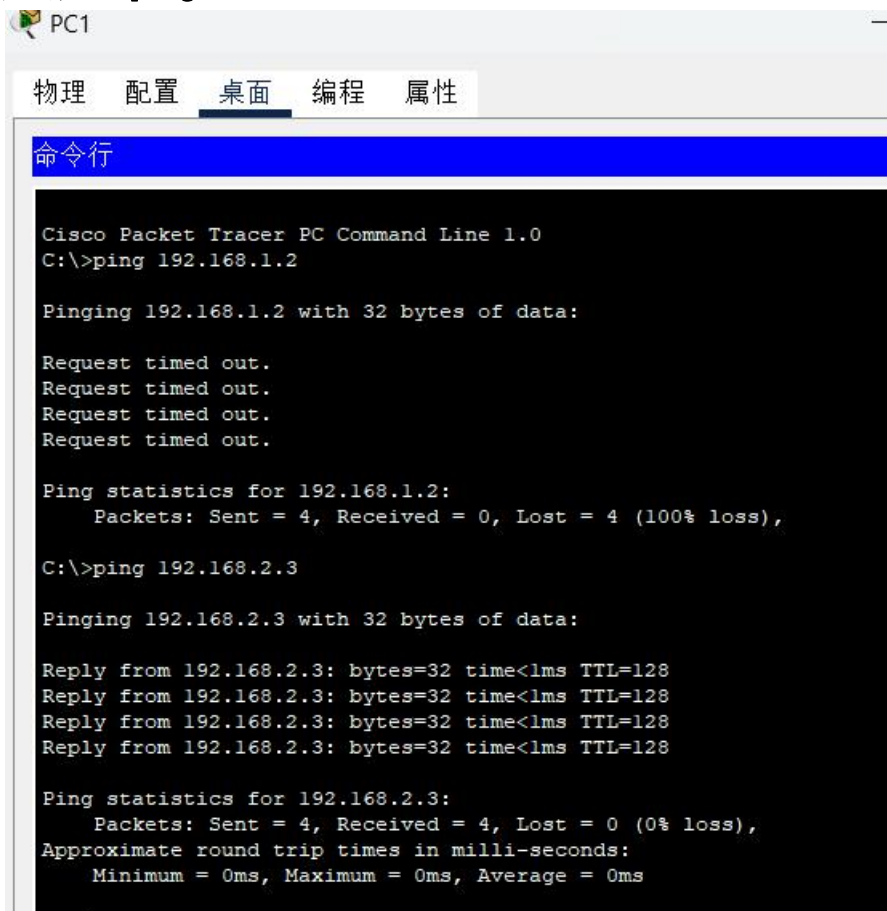


图 9 PC1 ping 192.168.1.2, 192.168.2.3 发现同一网段的能够 ping 通, 不同网段的无法 ping 通

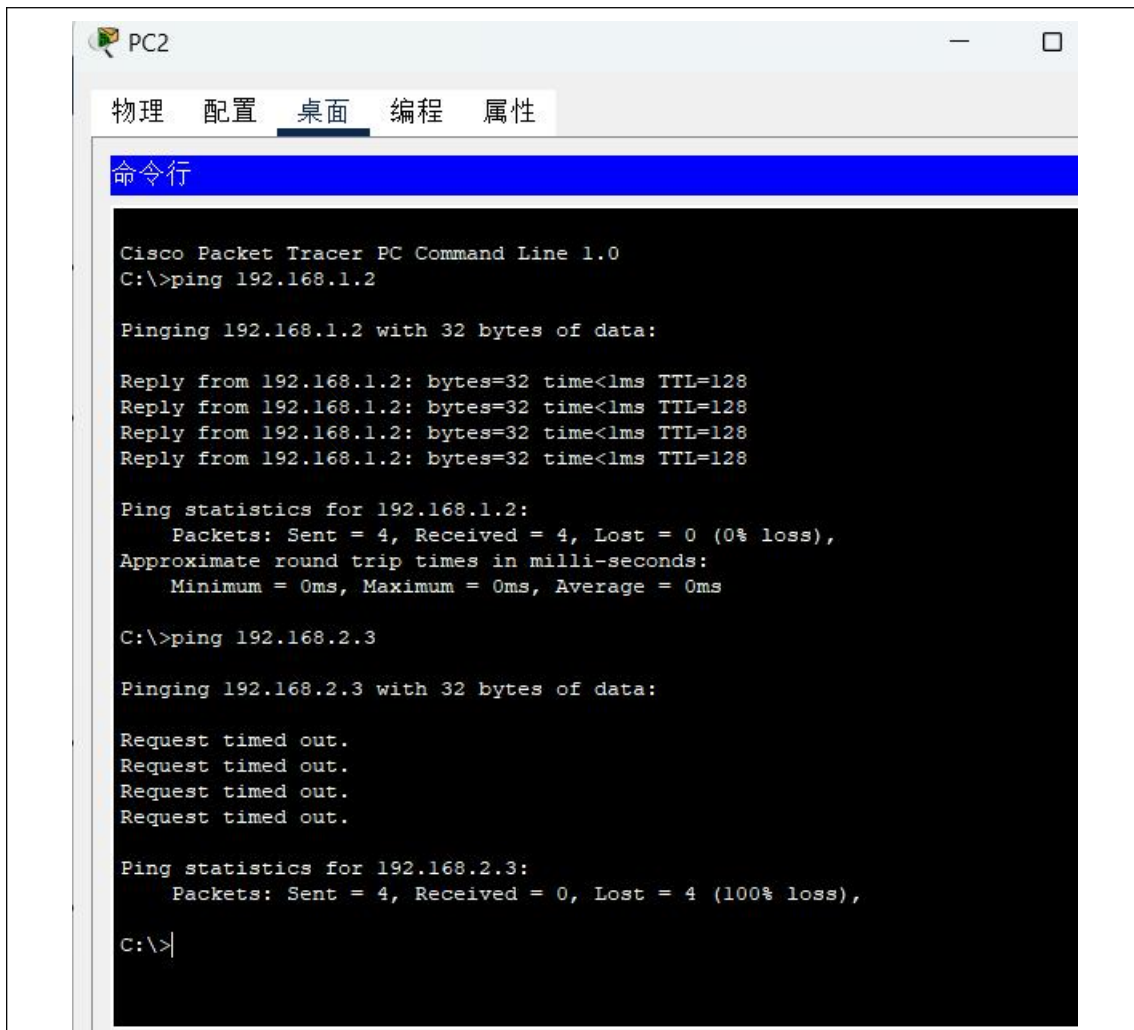


图 10 PC2 ping 192.168.1.2, 192.168.2.3 发现同一网段的能够 ping 通, 不同网段的无法 ping 通

二层交换机 2960 的配置命令

```

Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#exit
Switch(config)#inter
Switch(config)#interface fa0/2
Switch(config-if)#switc
Switch(config-if)#switchport acc
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fa0/3
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#intre
Switch(config)#inter
Switch(config)#interface fa0/1
Switch(config-if)#switc
Switch(config-if)#switchport mode tru
Switch(config-if)#switchport mode trunk

```

图 11 进入全局配置模式，创建了 vlan10、vlan20 两个 vlan，进入连接 PC 机的端口，将该端口分配到对应的 vlan，进入连接三层交换机的端口，将端口设置为 trunk 模式

三层交换机 3560 的配置命令

```
Switch>ena
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#exit
Switch(config)#inter
Switch(config)#interface fa0/1
Switch(config-if)#switc
Switch(config-if)#switchport tr
Switch(config-if)#switchport trunk en
Switch(config-if)#switchport trunk encapsulation do
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#swit
Switch(config-if)#switchport mode tr
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#interface fa0/2
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
```

图 12 进入全局配置模式，创建了 vlan10、vlan20 两个 vlan，然后进入连接二层交换机的端口，将 trunk 封装设置为 802.1q 协议，然后将端口设置为主干道 Trunk 模式

```
Switch(config)#inter
Switch(config)#interface vlan 10
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up

Switch(config-if)#ip address 192.168.1.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 20
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up

Switch(config-if)#ip address 192.168.2.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#ip routing
Switch(config)#
```

图 13 进入 vlan 的配置模式，为 vlan 接口配置 IP 地址和子网掩码，然后开启该端口，最后使用 ip routing 命令可以开启三层交换机的 IP 路由功能。

配置交换机 3560 后 PC 之间的连通性测试结果图

```

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<lms TTL=127
Reply from 192.168.2.2: bytes=32 time<lms TTL=127
Reply from 192.168.2.2: bytes=32 time<lms TTL=127
Reply from 192.168.2.2: bytes=32 time<lms TTL=127

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time<lms TTL=127
Reply from 192.168.2.3: bytes=32 time<lms TTL=127
Reply from 192.168.2.3: bytes=32 time<lms TTL=127
Reply from 192.168.2.3: bytes=32 time<lms TTL=127

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

图 14 PC0 ping 192.168.2.2 和 192.168.2.3 发现都能 ping 通

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<lms TTL=127
Reply from 192.168.1.2: bytes=32 time<lms TTL=127
Reply from 192.168.1.2: bytes=32 time<lms TTL=127
Reply from 192.168.1.2: bytes=32 time<lms TTL=127

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<lms TTL=128
Reply from 192.168.2.2: bytes=32 time<lms TTL=128
Reply from 192.168.2.2: bytes=32 time<lms TTL=128
Reply from 192.168.2.2: bytes=32 time<lms TTL=128

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

```

图 15 PC3 ping 192.168.1.2 和 192.168.2.2 也能够 ping 通

4、总结

通过本次实验更深刻地理解 VLAN 在网络设计中的重要性，包括如何有效隔离广播域，提高网络的安全性和性能。亲手配置 VLAN，从而更好地掌握了如何在真实网络环境中创建 VLAN、分配端口以及配置主干。

加深了 Trunk 端口和 Switched Virtual Interface (SVI) 的理解，通过设置 `switchport mode trunk` 和 `switchport trunk encapsulation dot1q` 等指令，成功实现了允许多个 VLAN 的数据通过同一个物理链接传输，而不会造成信息的混乱或丢失。

Switched Virtual Interface (SVI) 的配置为我提供了 VLAN 间路由的实践经验。通过为每个 VLAN 创建一个 SVI 并赋予它一个 IP 地址，并学会了如何配置它们以实现 VLAN 间的通信。

在实验的过程中，我也意识到了精确的网络规划和设计的必要性。每一步的配置都需要提前规划，包括对于网络拓扑的设计、选择合适的 VLAN 划分方案以及适当地设置主干链接和 SVI。这些规划对于确保网络不仅在当前高效运作，而且在未来易于扩展且安全可靠至关重要。

实验成绩评定表

序号	考核项目	分值分布	成绩
1	出勤与纪律情况	10	
2	实验任务完成情况	40	
3	实验报告质量	50	
总分			
指导教师签字			

备注：考核项目及分值分布应依据教学大纲确定。